TravelTide

Rewards Program

Customer Insights & Perk Assignment Strategy

Presented to: Elena Tarrant, Head of Marketing

Analyst: Jason Zachary Guest | April 2026

Agenda

01 Company & Project Timeline
Where TravelTide started and how we approached the data

02 Problem & Scope
What Elena asked for and the analytical constraints we worked within

03 Data Foundation
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04 Demographics
Who our customers are: age, family status, geography

05 Customer Lifecycle
The biggest pattern in the data: 5 tiers of platform engagement

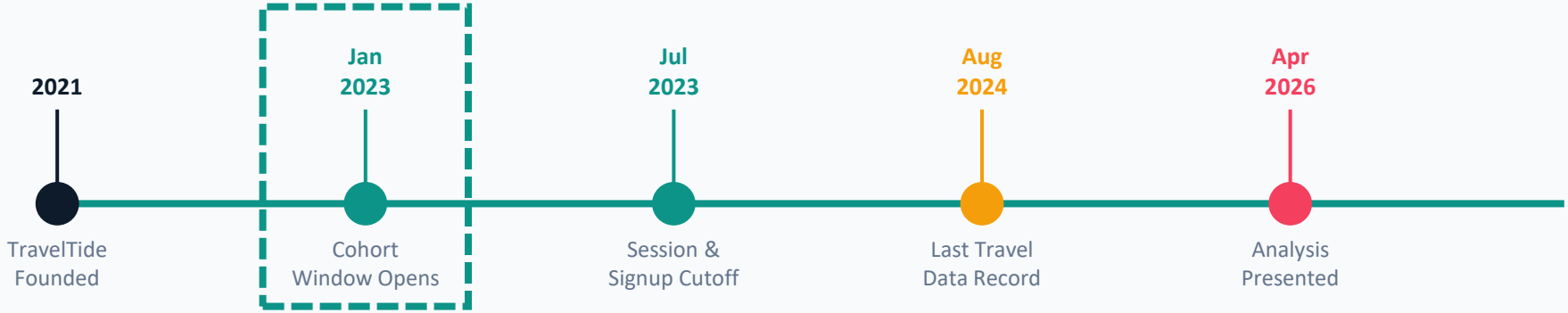
06 Booking & Spending Patterns
How customers book, spend, cancel, and use discounts

07 Perk Strategy
How we assigned 5 perks to 5,998 users using behavioral logic

08 Recommendations & Next Steps
A/B testing, loyalty program, and platform improvements

Company & Project Timeline

From TravelTide's founding to today's analysis



Note: All age calculations use **August 19, 2024** as the reference date (last recorded travel date).
Current date analysis is intentionally excluded to maintain data integrity across all relative calculations.

Average days since signup across ALL cohort tiers ≈ 570 days
meaning everyone in this cohort had approximately the same opportunity to engage with the platform.

Problem & Scope

What Elena asked for — and how we defined the boundaries

The Ask

Elena Tarrant (Head of Marketing) needs to move from sending generic promotional emails to personalized reward offers — showing each customer the ONE perk they are most likely to respond to.

The 5 proposed perks*:

- 💰 Exclusive Discounts
- 🔄 No Change Fees
- ✈️ Free Checked Bags
- 🏨 Free Hotel Meals
- ⭐ Premium Travel Benefits

** Add-on Loyalty program perk also discussed*

Analytical Constraints

Cohort Definition:
- Active users with 7+ sessions since Jan 4, 2023

Mutually Exclusive:
- Exactly ONE perk per user - no overlaps

Audience:
- Non-technical senior leadership presentation

Lifecycle Gating:
- Minimum 3 trips required for full behavioral analysis

Data Anchor:
- All calculations relative to Aug 19, 2024

Geography:
- North American cohort only — USA & Canada

Data Foundation

From raw sessions to 160+ behavioral data points per user

49,218
Total Sessions in Cohort

5,998
Cohort Users Analyzed

~50
Original Data Points / User

~160
Engineered Features / User

Raw Sessions
50K sessions
6K users

→

Cohort Filtering
7+ sessions
since Jan 2023

→

Data Cleaning
Session types
Anomaly fixes

→

Feature Engineering
5-table
architecture

→

ML Clustering
K-Means
6 dimensions

→

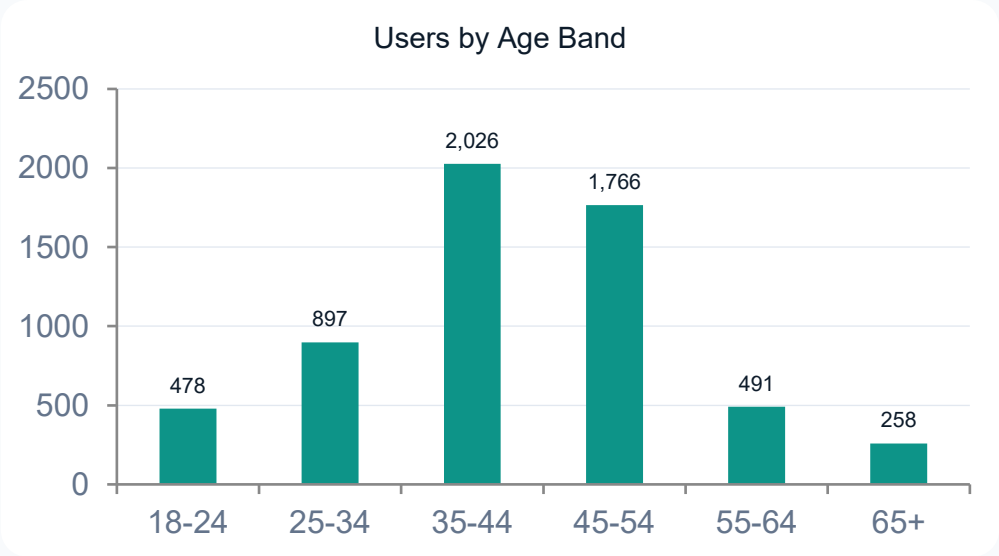
Perk Assignment
5,998 users
1 perk each

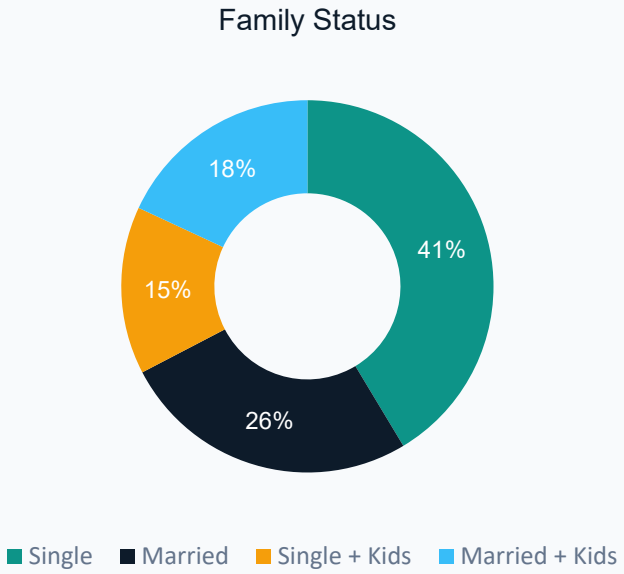
Key Anomalies Discovered & Resolved During Analysis

- Cancel sessions showed avg duration of 5,509 sec (56.6% hit 2-hr timeout) | session duration excluded from cancel analysis; flagged for IT/website team
- Flight-only avg lead time = 44.9 days but median = 7 days | 14% of flight bookings placed 180+ days out skew the mean; median used throughout
- Discount rates >1.0 found in session-level data | resolved via trip-level deduplication; 4 deprecated airport codes added to reference table

Who Are Our Customers?

Demographics — Age, Family Status & Geography





Core customer: 35-54 age band = 63.2% of cohort, highest spend ($3,682-$3,691 avg), highest booking conversion (38%).

Geography: USA 82.7% | Canada 17.3% | EMEA expansion would require cohort reanalysis

The Biggest Pattern We Found

Customer Lifecycle

	Tier 0	Tier 1	Tier 2	Tier 3	Tier 4
	Never Booked	Single Trip	Two Trips	Established 3-5 Trips	Loyal 6-9 Trips
	522 users	939 users	1,315 users	2,965 users	257 users
	8.7% of cohort	15.7% of cohort	21.9% of cohort	49.4% of cohort	4.3% of cohort
Avg Spend	$0	$1,344	$2,595	$4,235	$6,410
Iter 1 10% up		+$70K 52 users → Tier 1	+$118K 94 users → Tier 2	+$216K 132 users → Tier 3	+$646K 297 users → Tier 4

** ~570 days since signup across all tiers - every user had the same window to grow*

Iteration 1 Total Uplift: +$1,049,937

Lifecycle Tier — Detailed Metrics

Spend, cancellation, booking type mix and engagement by tier

Lifecycle Tier	Users	Total Revenue	Avg Spend	Days to 1st Trip	Cancel Rate	% Package	% Hotel Only	Trips/Year
Tier 0 — Never Booked	522	$0	$0	—	100%	—	—	0.0
Tier 1 — Single Trip	939	$1,218,638	$1,344	53 days	54.5%	64.9%	31.8%	2.9
Tier 2 — Two Trips	1,315	$3,407,035	$2,595	21 days	35.8%	70.3%	20.1%	5.5
Tier 3 — Established	2,965	$12,556,581	$4,235	8 days	25.1%	74.8%	12.8%	9.0
Tier 4 — Loyal	257	$1,647,334	$6,410	2 days	17.1%	73.8%	9.2%	12.7
TOTAL COHORT	**5,998**	**$18,829,588**	**$3,139**					

Booking & Spending Patterns

How customers book, what they spend, and how they use discounts

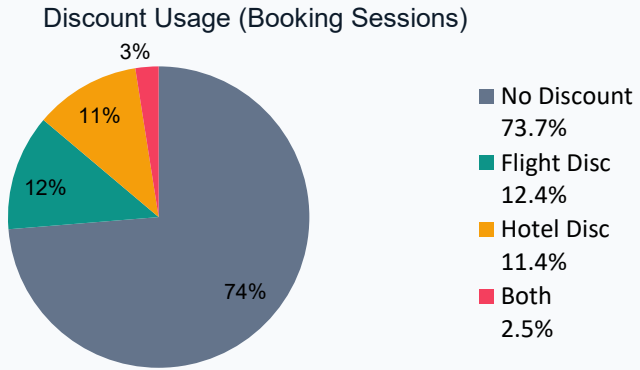
Based on 16,099 Booking sessions
Out of 49,218 total sessions

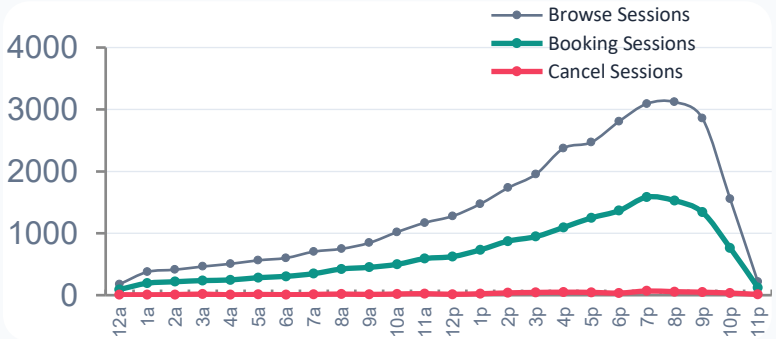
74.1%
of bookings are packages (flight+hotel)

73.7%
of bookings use no discount at all

$161.98
avg discount value when both discounts applied

44.9 days
avg flight-only lead time (median = 7 days)*





Peak Booking: 4PM - 9PM | **Single highest:** 7PM–8PM (1,584 bookings)

** Flight-only lead time is mean-skewed by 14% of bookings placed 180+ days in advance. Always use median for reporting.*

Flight Travel Patterns

International vs Domestic — fare premium, distance and checked bags

Metric	International	Domestic	Variance
Flight Bookings	5,269 (38.4%)	8,448 (61.6%)	—
Avg Gross Fare	$614.50	$413.49	**+49% Intl**
Avg Fare per Seat	$452.99	$349.66	**+30% Intl**
Avg Distance (km)	2,601.9	2,000.0	+30% Intl
Avg Checked Bags	0.596	0.574	Similar
Avg Lead Time (days)	22.4	10.8	**+107% Intl**

38.4%
of all flights are international

+49%
fare premium for international travel

22 days
intl avg lead time vs 11 days domestic

International travelers book further in advance, pay significantly higher fares, and represent a premium retention-critical segment.

Perk Assignment Strategy

Exclusive Discounts	No Change Fees	Free Checked Bags	Free Hotel Meals	Premium Travel Benefits
2,546	**233**	**715**	**221**	**2,283**
users 42.4%	users 3.9%	users 11.9%	users 3.7%	users 38.1%
Avg Spend	Avg Spend	Avg Spend	Avg Spend	Avg Spend
$2,332	**$1,768**	**$3,692**	**$3,987**	**$4,490**
Key Signal	Key Signal	Key Signal	Key Signal	Key Signal
Never booked or < 3 trips default	Cancel spend ratio ≥ 40%	≥ 90% intl flights completed	Hotel discount rate ≥ 50%	High freq, high spend low discount use

Total cohort revenue: $18,829,588 | 5,998 users assigned | 0 unassigned

How Perks Are Assigned

Lifecycle-gated behavioral logic - dynamic, not static

Tier 0 Never Booked 522 users	**Exclusive Discounts**	Primary objective - convert first booking Never booked = no relationship
Tier 1 1 Trip 939 users	**No Change Fees or Exclusive Discounts**	1st trip canceled → No Change Fees Otherwise → continue Exclusive Discounts
Tier 2 2 Trips 1,315 users	**3 options available**	100% international travel → Free Checked Bags >=1 trip cancel → No Change Fees Otherwise → continue Exclusive Discounts
Tier 3+ 3+ Trips 3,222 users	**All 5 perks eligible**	Full behavioral signal evaluation applies

Perks are dynamic - reassignment triggered as users accumulate trips. A never-booked user today can become a Premium Travel Benefits user tomorrow.

Recommendations & Next Steps

Four actions to validate and grow the rewards program

01 A/B Testing - 30 Day Pilot

Randomly sample: 50% of cohort users per perk assignment.

- Test group receives 1 personalized perk email/week.
- Control group receives standard generic promotional email.

Measure:

- booking conversion rate, cancellation rate, revenue per user.
- Avg booking lead time < 2 weeks = results within 30 days.

02 Loyalty Tier Program

- **Formalize the lifecycle tiers** as a visible customer loyalty program based on completed (non-cancelled) trips.
- Tier progression rewards create a self-reinforcing conversion incentive.
- Progressing 10% of existing platform users two tiers would generate an **estimated $1.8M increase** in platform revenue.

03 Cancellation UX Review

- 75.9% of **cancel sessions exceed 1 hour**
- 56.6% hit the **2-hour system timeout exactly**.
- The cancellation flow does not actively terminate sessions on completion.
- Recommend IT/website team review: implement session-close on cancellation confirmation. Also review flight-only cancellation rate - disproportionately high vs booking share.

04 Perk Threshold Review Cycle

- **Perk thresholds are data-driven** but should be reviewed quarterly as user behavior evolves.

Key triggers for a review:

- >15% users unassigned after all rules applied; cohort expands to EMEA; cancel spend ratio baseline shifts significantly.
- The 80% intl threshold (34 additional users) is the first adjustment candidate for Free Checked Bags.

Thank You

+$1,780,927 projected revenue uplift if 10% of users in each lifecycle tier progress just two tiers up — using the behavioral model built in this analysis.

5,998	users - every one assigned exactly one perk
$18.8M	total cohort revenue profiled and segmented
0	users unassigned = complete coverage
5	perks - mapped to real behavioral signals, not assumptions

The decision for today?
Approve the A/B test. 30 days. 5 emails. Measure results.

Questions?